

HTTP Basics:

The screenshot shows a web browser window with the URL `127.0.0.1:8080/WebGoat/start.mvc#lesson/HttpBasics.lesson/1`. The page title is "HTTP Basics". On the left, there is a sidebar menu with the following items: Introduction, General, HTTP Basics (highlighted), HTTP Proxies, Developer Tools, CIA Triad, Crypto Basics, Writing new lesson, (A1) Injection, (A2) Broken Authentication, (A3) Sensitive Data Exposure, (A4) XML External Entities (XXE), (A5) Broken Access Control, (A7) Cross-Site Scripting (XSS), (A8) Insecure Deserialization, (A9) Vulnerable Components, (A8:2013) Request Forgeries, Client side, and Challenges. The main content area has a "Show hints" button and a "Reset lesson" button. Below these are three numbered steps: 1 (blue), 2 (green), and 3 (red). Step 2 is the active step. The text for step 2 reads: "Enter your name in the input field below and press 'Go!' to submit. The server will accept the request, reverse the input and display it back to the user, illustrating the basics of handling an HTTP request." Below this text is a "Try It!" section. It contains a text input field with the value "Lake" and a "Go!" button. Below the input field, it says "The server has reversed your name: ekaL".

WebGoat

127.0.0.1:8080/WebGoat/start.mvc#lesson/HttpBasics.lesson/1

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WEBGOAT

HTTP Basics

Introduction >

General >

HTTP Basics

HTTP Proxies

Developer Tools

CIA Triad

Crypto Basics

Writing new lesson

(A1) Injection >

(A2) Broken Authentication >

(A3) Sensitive Data Exposure >

(A4) XML External Entities (XXE) >

(A5) Broken Access Control >

(A7) Cross-Site Scripting (XSS) >

(A8) Insecure Deserialization >

(A9) Vulnerable Components >

(A8:2013) Request Forgeries >

Client side >

Challenges >

Show hints Reset lesson

1 2 3

Enter your name in the input field below and press "Go!" to submit. The server will accept the request, reverse the input and display it back to the user, illustrating the basics of handling an HTTP request.

Try It!

Enter your name in the input field below and press "Go!" to submit. The server will accept the request, reverse the input and display it back to the user, illustrating the basics of handling an HTTP request.

Enter Your Name: Lake Go!

The server has reversed your name: ekaL

HTTP Proxies:

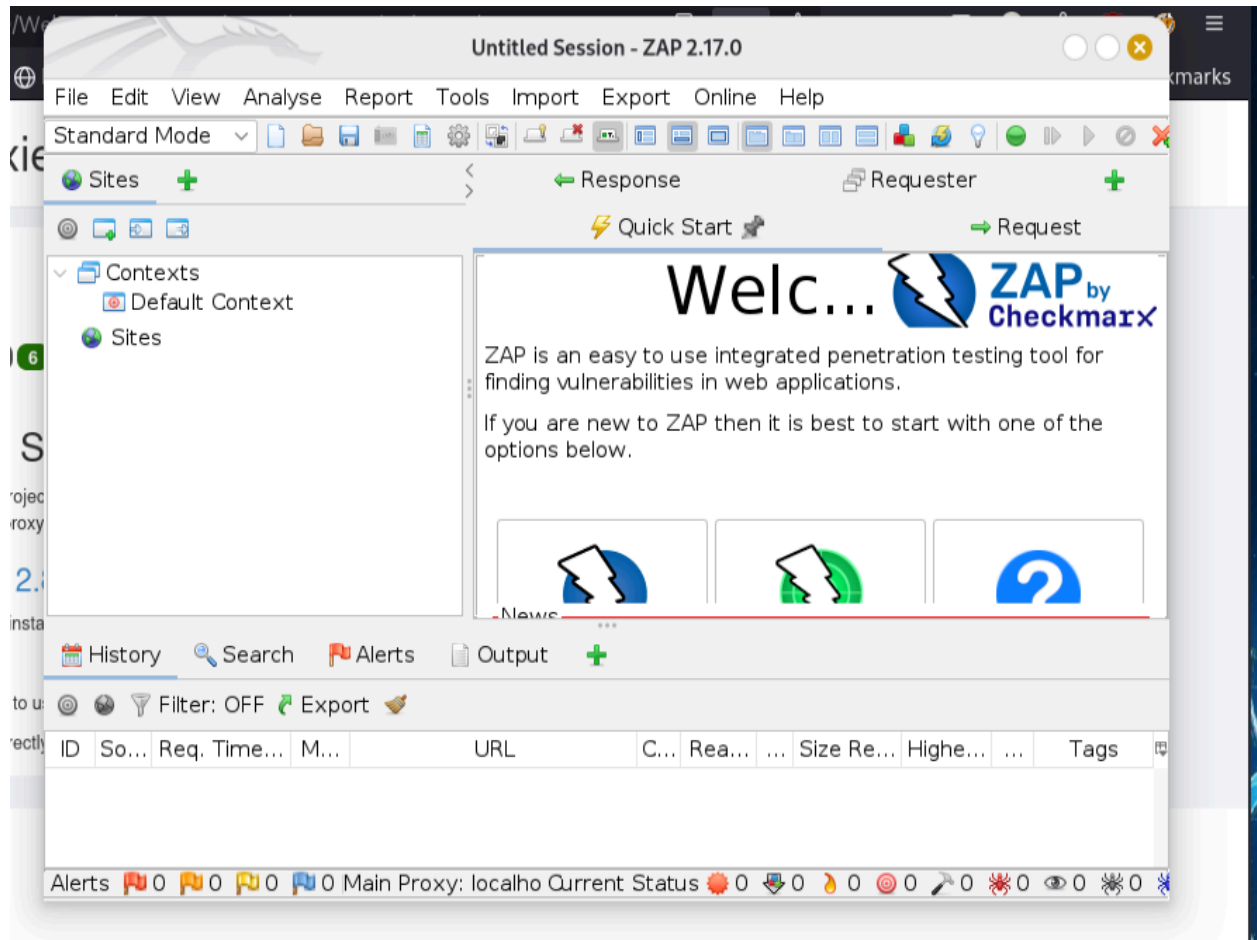
Installed ZAP in the command line

```
[redacted]-[~/Desktop]
$ sudo apt install zaproxy
[sudo] password for [redacted]
Installing:
  zaproxy

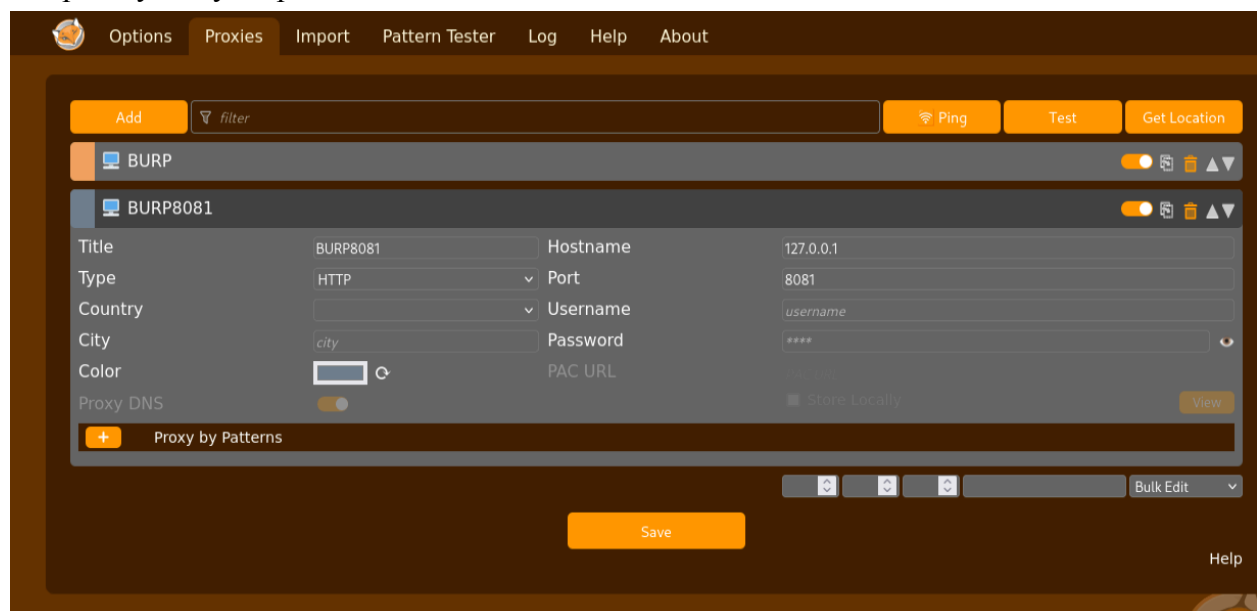
Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 2390
  Download size: 222 MB
  Space needed: 280 MB / 8,409 MB available

Get:1 http://kali.download/kali kali-rolling/main amd64 zaproxy all 2.17.0-0kali1 [222 MB]
Fetched 222 MB in 14s (16.2 MB/s)
Selecting previously unselected package zaproxy.
(Reading database ... 442092 files and directories currently installed.)
Preparing to unpack .../zaproxy_2.17.0-0kali1_all.deb ...
Unpacking zaproxy (2.17.0-0kali1) ...
Setting up zaproxy (2.17.0-0kali1) ...
Processing triggers for kali-menu (2025.2.7) ...
```

Launched ZAP and set the port to 8081



Setup FoxyProxy on port 8081



The screenshot shows the ZAP 2.17.0 interface. A 'Filter History' dialog box is open, displaying a list of filters. The dialog box has a title bar with 'Filter History' and standard window controls. Below the title bar, there is a text area with instructions: 'Select the required filters below. You can select multiple rows in each element. An element is not used for filtering if none of the rows in it are selected.' Below this, there are five columns: 'Methods:', 'Codes:', 'Tags:', 'Alerts:', and 'URL Inc Regex:'. Each column contains a list of items. The 'Methods' column lists HTTP methods: CONNECT, DELETE, GET, HEAD, OPTIONS, PATCH, POST, PUT, and TRACE. The 'Codes' column lists status codes: 100, 101, 200, 201, 202, 203, 204, 205, and 206. The 'Tags' column lists various tags: Comment, Form, JSON, MailTo, Password, Script, and SetCookie. The 'Alerts' column lists alert levels: Information, Low, Medium, High, False Pos, Low, and Medium. The 'URL Inc Regex' column contains two regex patterns: '.*WebGoat.*' and '.*lesson.*.mvmc'. Below the columns, there is a 'Notes' dropdown menu set to 'Ignore'. At the bottom of the dialog box, there are three buttons: 'Cancel', 'Clear', and 'Apply'. The background shows the ZAP main window with a 'Welcome to ZAP' message and a list of recent requests. The 'Sites' pane on the left shows a list of sites, including 'http://127.0.0.1:8080'. The 'History' pane at the bottom shows a list of requests with columns for ID, So..., Req. Time, Notes, and various other details.

File Edit View Analyse Report Tools Import Export Online Help

Standard Mode

Sites + Quick Start Request Response Requester +

Contexts
Default Context
Sites
http://127.0.0.1

Welcome to ZAP



ZAP is an easy to use integrated penetration testing tool for finding vulnerabilities in applications.

If you are new to ZAP then it is best to start with one of the options below.

Add Breakpoint

Location: Request Header
Match: Contains
String: POST
Inverse: ☐
Ignore Case: ☒

Save

Cancel

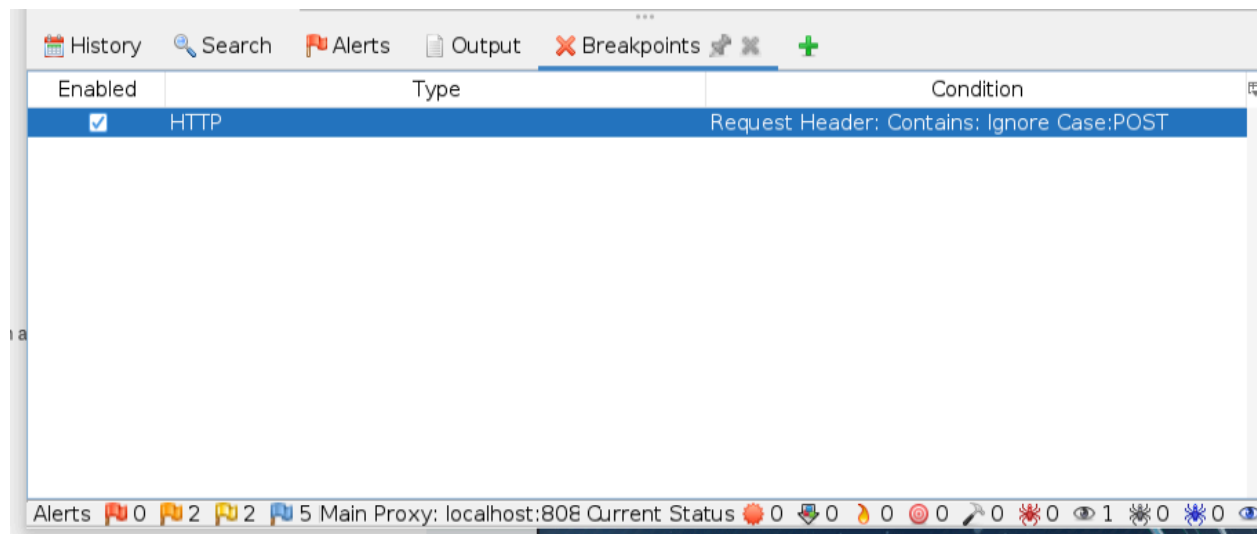


Learn More

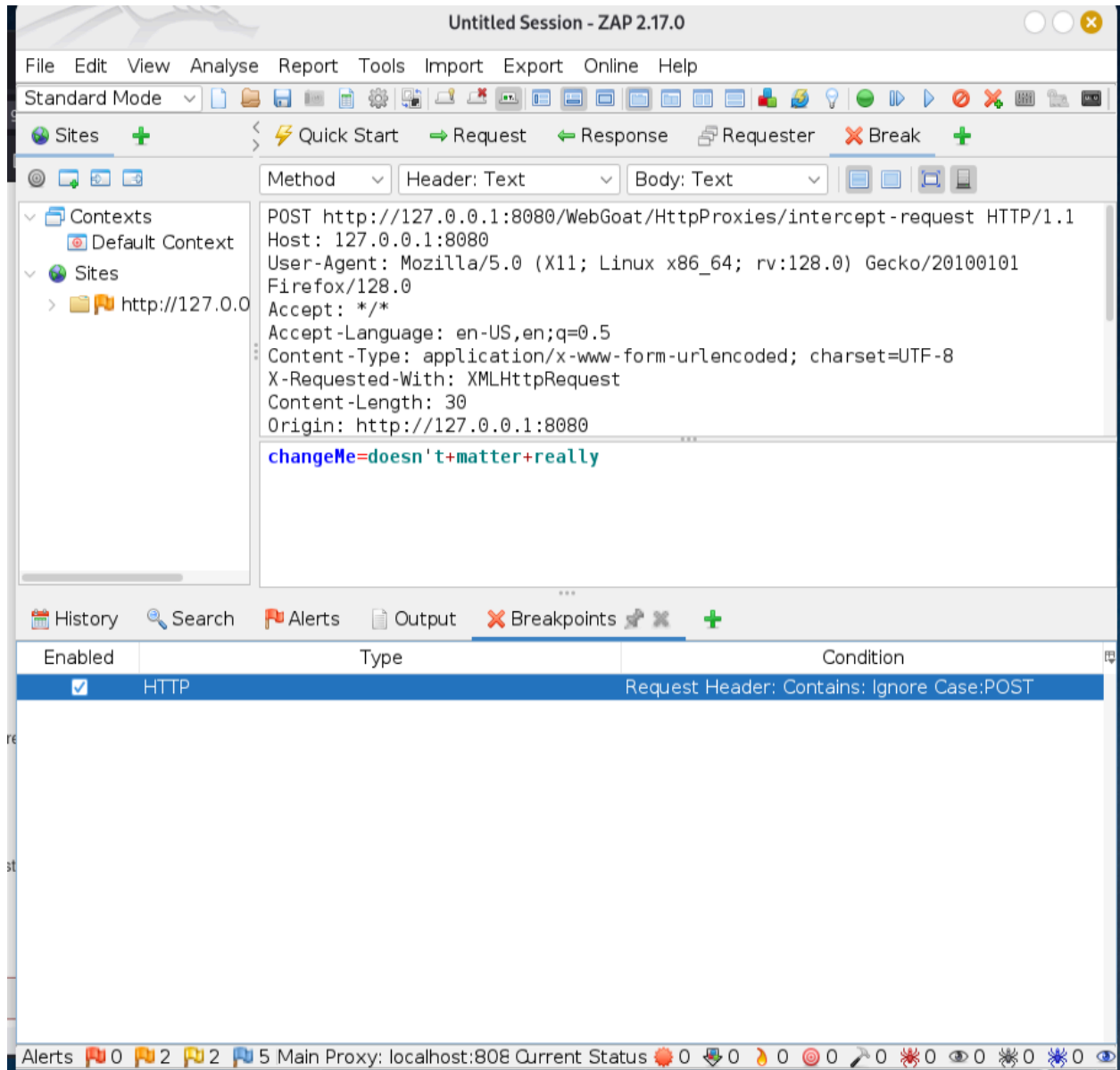
History Search Alerts

Filter: ON URL Inc Reg

ID	So...	Req. Timest...	Me...	URL	C...	Reason	...	Size Res...	Highes...	N...	T
241	...	2/17/26, 2:...	GEI	http://127.0.0.1:8080/Web...	2...	OK	...	532 bytes			
242	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	116 bytes			
243	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	863 bytes			
244	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	514 bytes			
245	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	252 bytes			
246	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	86 bytes			
247	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	48,513 ...			
248	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	1,393 b...	Medi...	Com	
252	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	3 bytes		JSON	
254	...	2/17/26, 2:...	GET	http://127.0.0.1:8080/Web...	2...	OK	...	22,292			



Intercept a request:



Modified the request

Submitted the request

The screenshot displays a web browser window on the left and a network tool on the right. The browser shows a POST request to `http://www.webgoat.local:8080/webGoat/httpBasics/attack2` with a body containing `magic_num=17&answer=GET&magic_answer=23`. The network tool on the right shows the intercepted request in the 'Request' tab, with the body text visible. Below the browser window, a text area contains instructions for intercepting and modifying a request, followed by a 'Submit' button. A green highlight is placed over the 'Submit' button and the text 'Well done, you tampered the request as expected'.

Method: POST
Header: Text
Body: Text

POST http://www.webgoat.local:8080/webGoat/httpBasics/attack2 HTTP/1.1
Host: www.webgoat.local:8080
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:78.0) Gecko/20100101 Firefox/78.0
Accept: */*
Accept-Language: nl,en-US;q=0.7,en;q=0.3
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
X-Requested-With: XMLHttpRequest
Content-Length: 39
Origin: https://www.webgoat.local:8080
Connection: keep-alive
Referer: https://www.webgoat.local:8080/webGoat/start.mvc
Cookie: JSESSIONID=ffrzov111uQ12KXp9uWYB-SjKQ1Dvxn2KCSlV
magic_num=17&answer=GET&magic_answer=23

Intercept and modify a request

Set up the intercept as noted above and then submit the form/request below by clicking the submit button. When you request the page, the request will be intercepted and you can modify it as follows.

- Change the Method to GET
- Add a header 'x-request-intercepted:true'
- Remove the request body and instead send 'changeMe' as query string parameter and set the value to 'Requests are intercepted'

Then let the request continue through (by hitting the play button).

The two play buttons behave a little differently, but we'll let you tinker and figure that out for yourself.

doesn't matter really [Submit]

Well done, you tampered the request as expected